

POWER BI: SUPERSTORE SALES DATA IMPORT, POWER QUERY & DATA CLEANING WEEK - 1 TASK

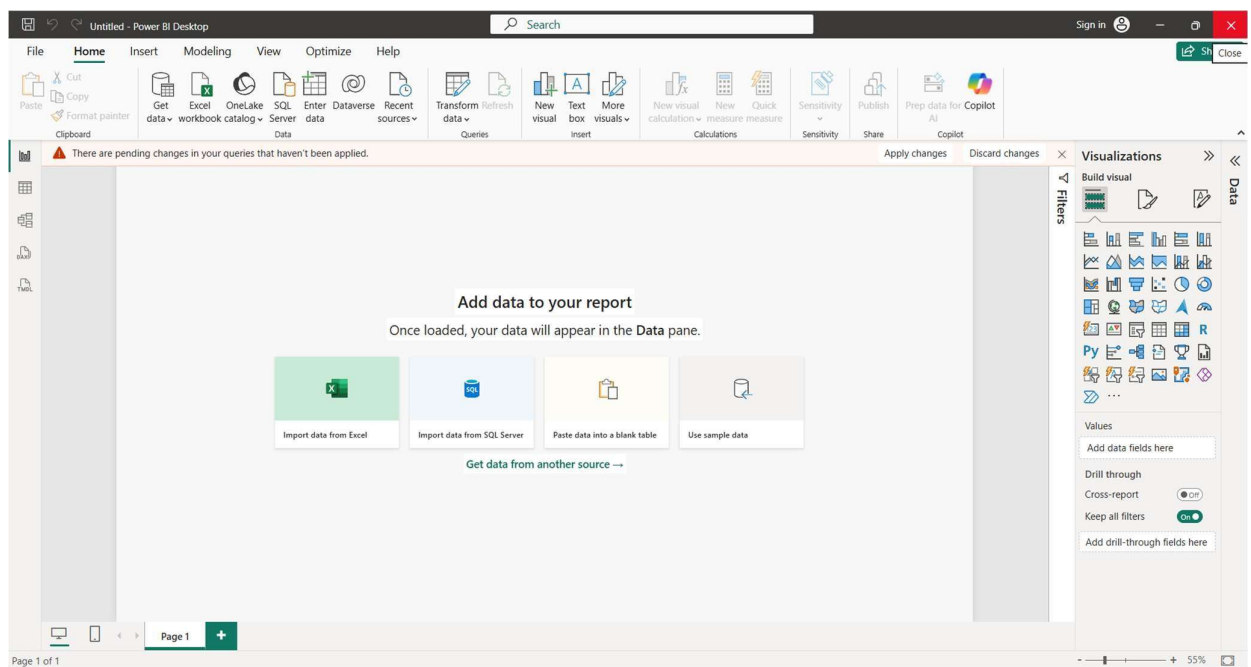
Name: Durga Devi S

Roll No: E25DA018

Course: B.Sc Data Science and Artificial
Intelligence

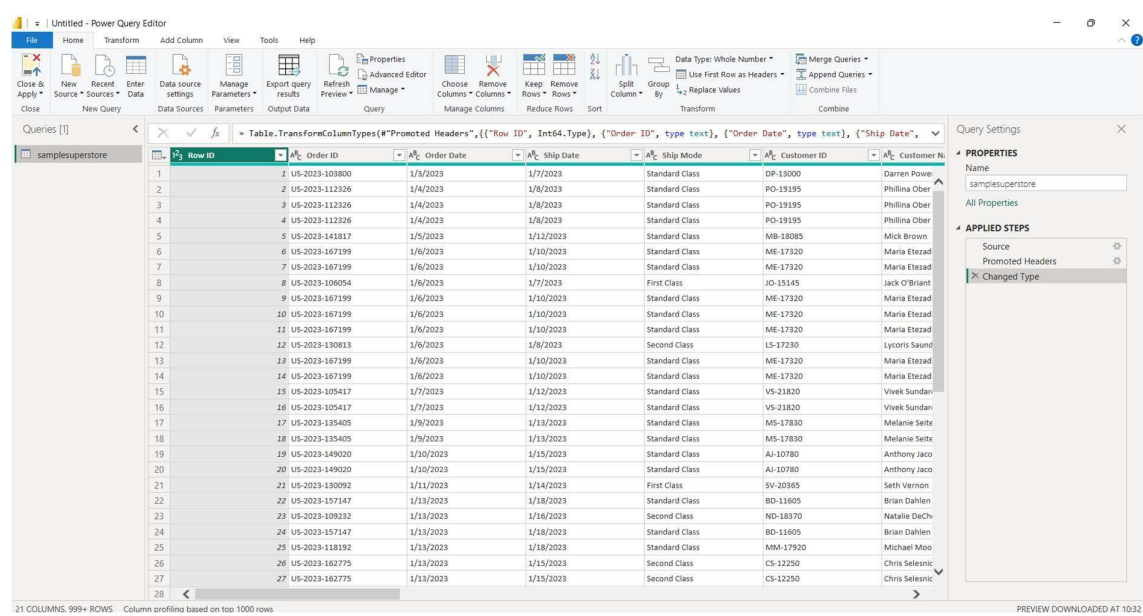
This week focuses on importing the Superstore Sales Dataset into Power BI Desktop and preparing the raw data for further analysis. The dataset is inspected and transformed using Power Query Editor by verifying data types, checking data quality, standardizing text fields, and extracting useful date attributes from the Order Date column. The required transformations are then applied to create a clean and structured dataset that can be used for data modeling and dashboard development in the following weeks.

Figure 1: Home Screen



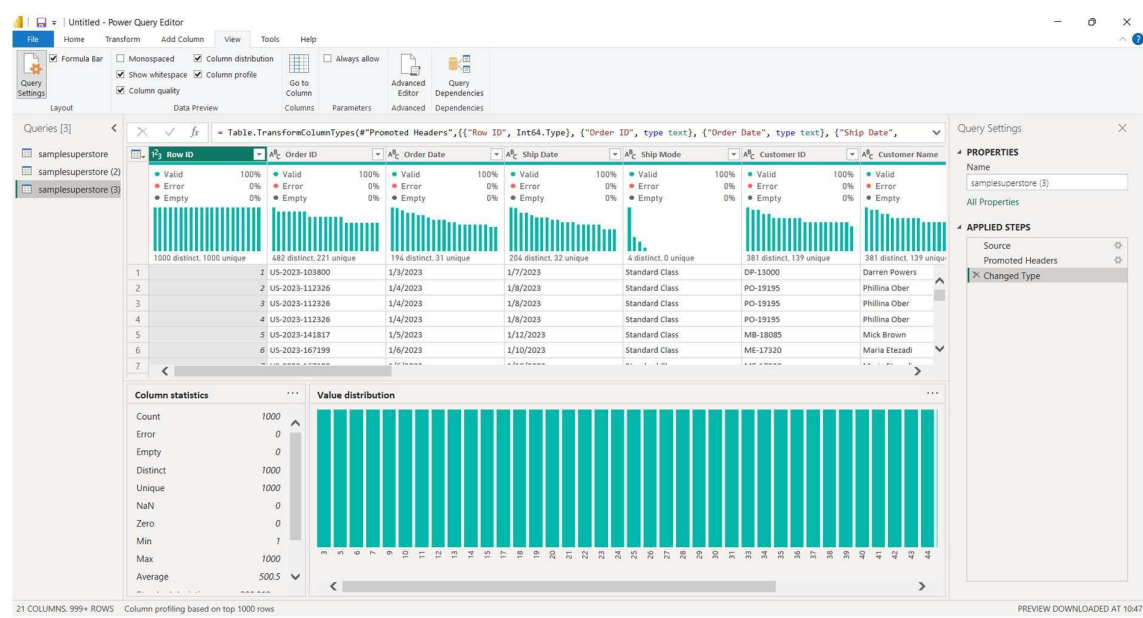
Home → Get Data → Text/CSV → select your Superstore CSV → Transform Data

Figure 2: Superstore Sales dataset imported and opened in Power Query Editor



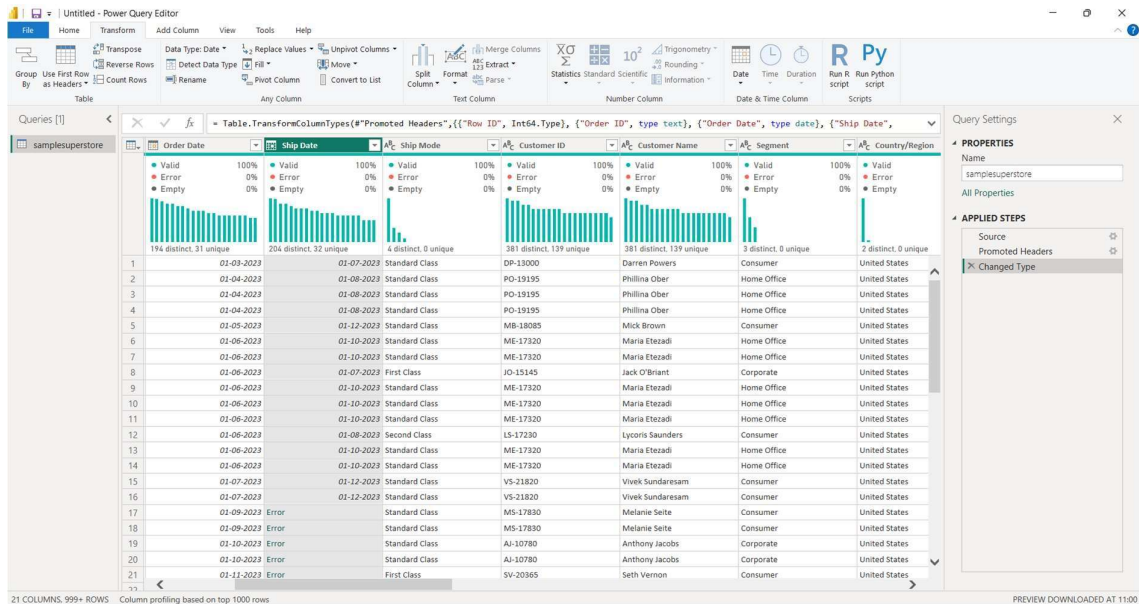
View → Data Preview → Column quality → Column distribution → Column profile → Turn On

Figure 3: Data quality inspection showing valid, error, and empty values



Order Date And Ship Date → Transform → Data Type → Date

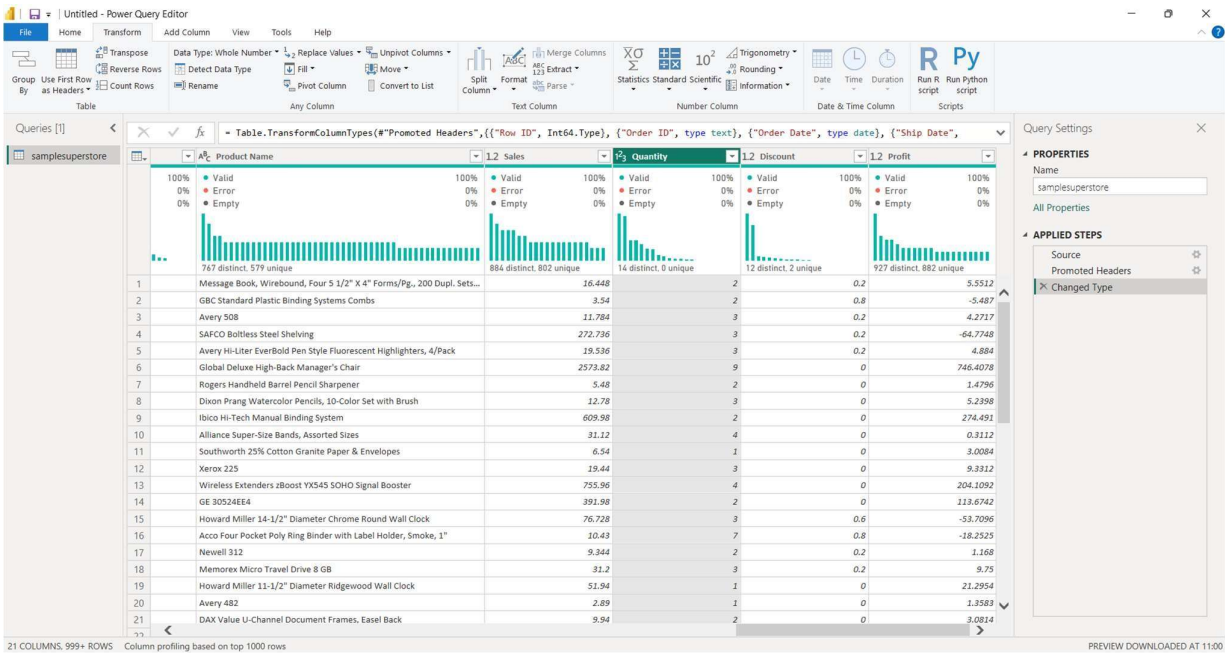
Figure 4: Order Date and Ship Date converted to Date data type in Power Query Editor



Select these columns

Sales → Quantity → Discount → Profit → Customer ID → Product ID
→ Order ID choose the
required data type

Figure 5: Numerical columns verified with appropriate data types in Power Query Editor



Trimming and Cleaning For:

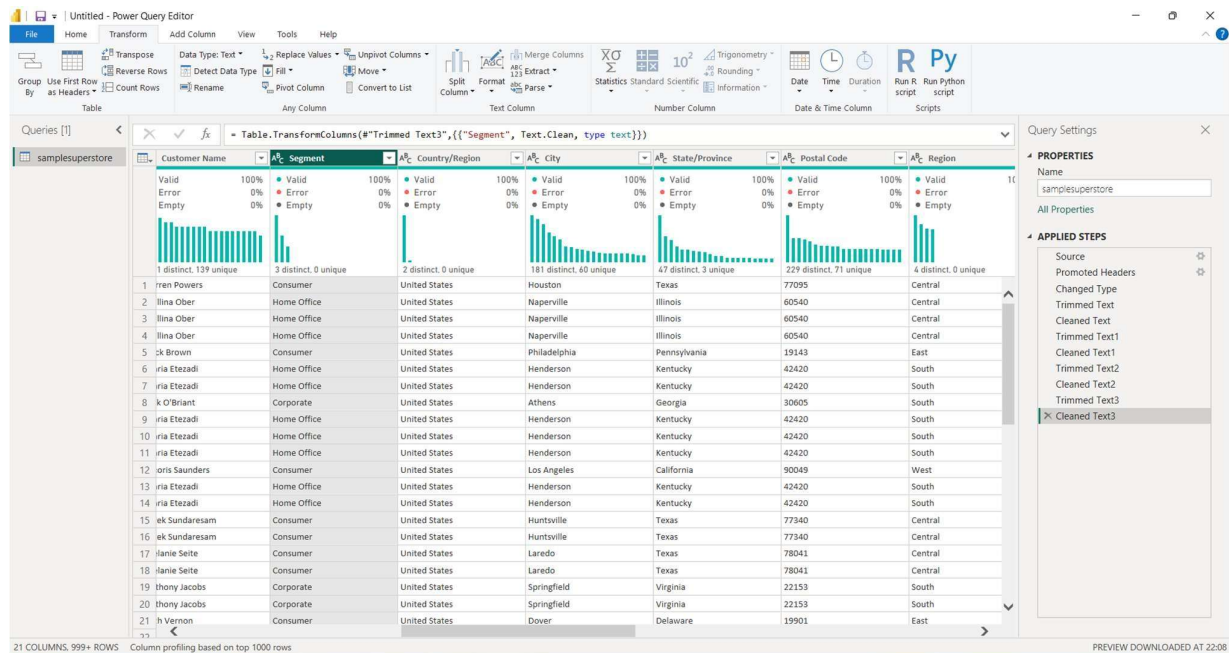
Category

Sub-Category

Segment

Region

Figure 6: Categorical fields standardized using Trim and Clean transformations in Power Query Editor



New Columns created from Order Date are:

Order Year

Order Quarter Number

Order Quarter Order Month Number

Order Month

Order Day

Order Day of Week

Figure 7: Date attributes extracted from Order Date for temporal analysis

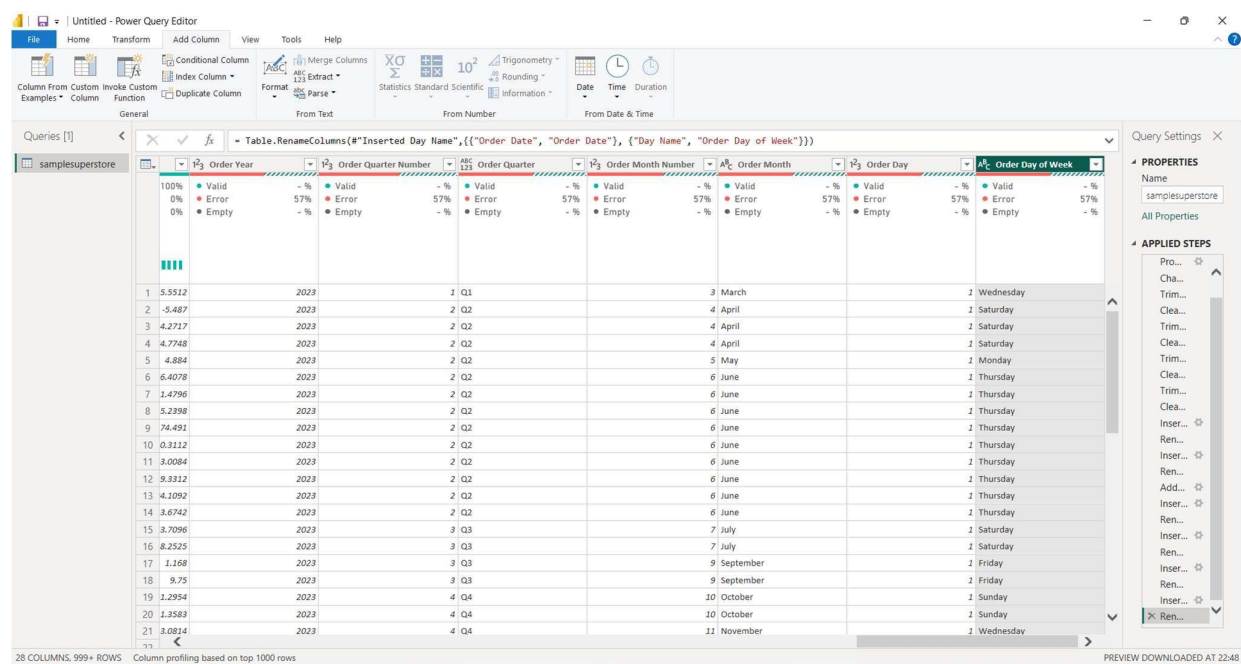


Figure 8: Applied Steps showing the sequence of data cleaning and transformation operations

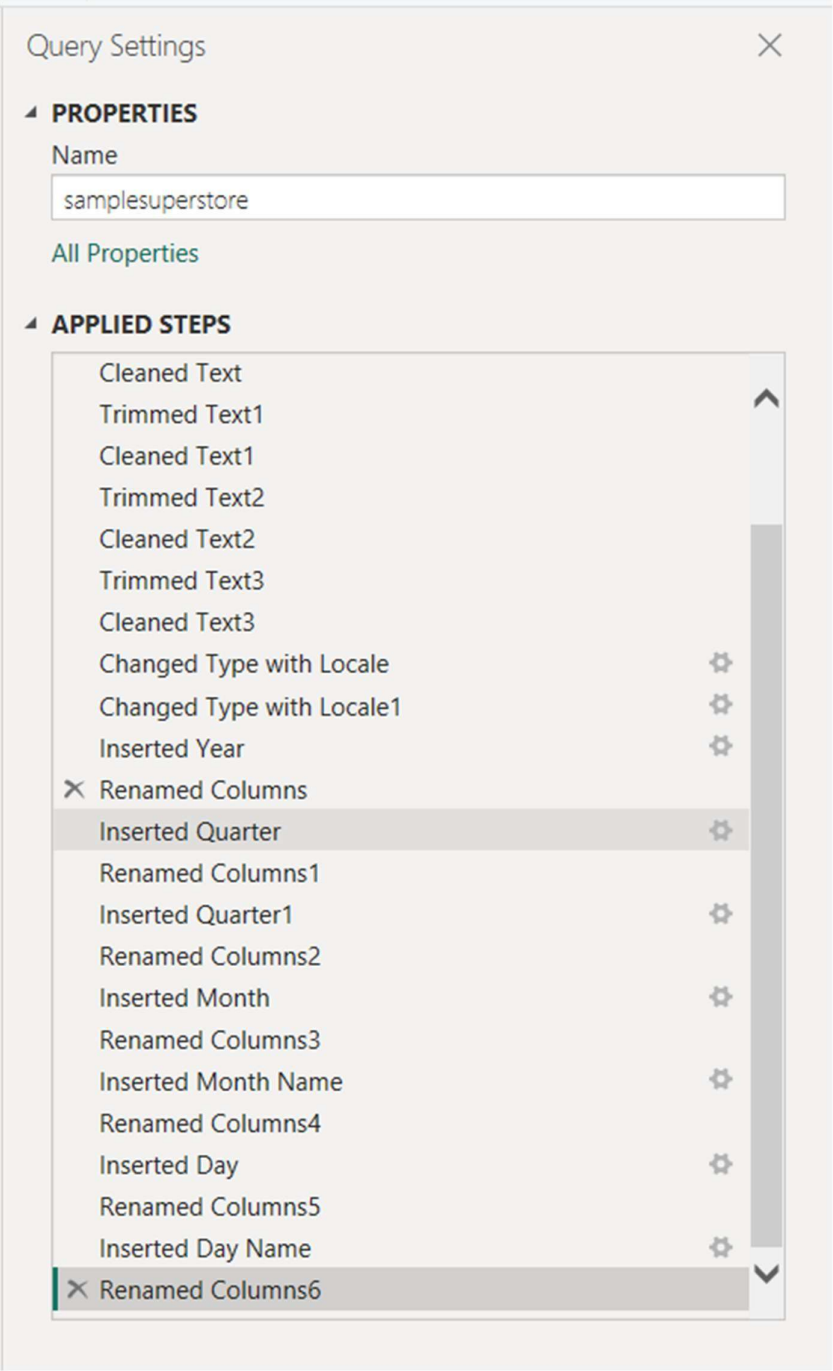
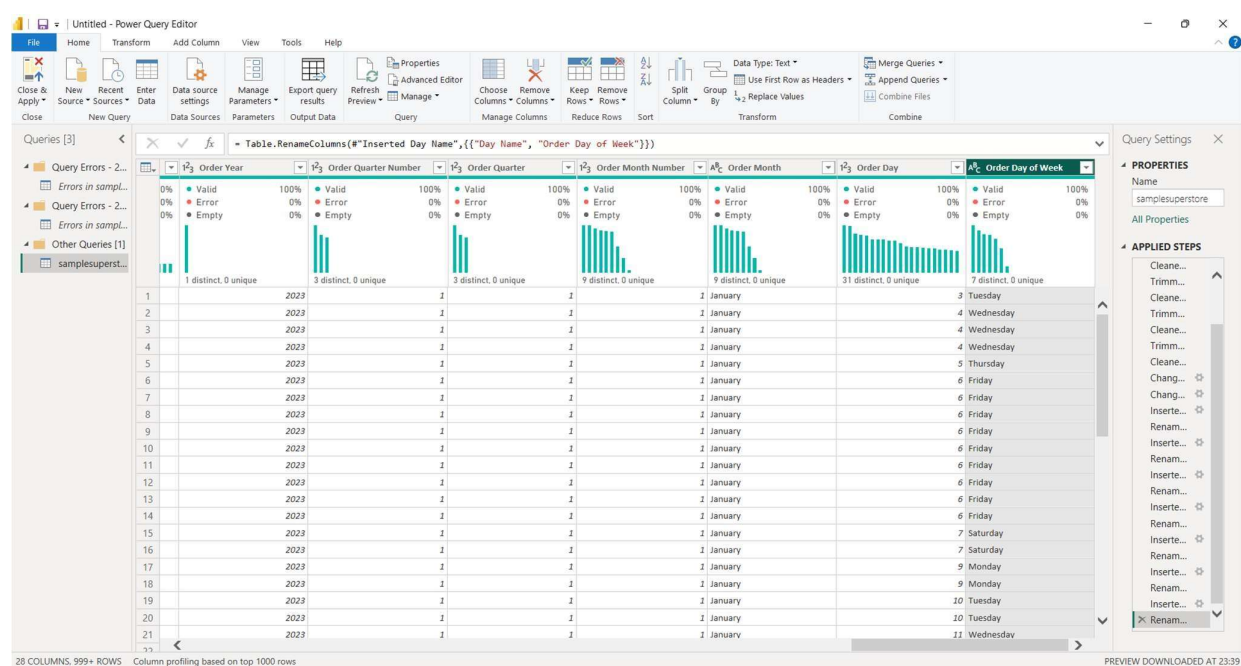


Figure 9: Final cleaned and transformed Superstore Sales dataset in Power Query Editor



Click Home

Close and Apply

Figure 10: Final transformed Superstore Sales dataset
verified in Data View

Table: samplesuperstore (10,194 rows)

Sales	Quantity	Discount	Profit	Order Year	Order Quarter Number	Order Month Number	Order Month	Order Day	Order Day of Week	Order Quarter
19.44	3	0	9.3312	2023	1	1	January	6	Friday	1
19.3	2	0	5.79	2023	1	1	January	20	Friday	1
19.36	2	0	9.2928	2023	1	1	January	20	Friday	1
12.96	2	0	6.2208	2023	1	2	February	20	Monday	1
91.96	2	0	15.6332	2023	2	4	April	6	Thursday	2
33.11	7	0	12.9129	2023	2	4	April	6	Thursday	2
55.48	1	0	26.6304	2023	2	4	April	6	Thursday	2
19.44	3	0	9.3312	2023	2	4	April	6	Thursday	2
187.76	4	0	76.9816	2023	2	4	April	15	Saturday	2
106.96	2	0	31.0184	2023	2	4	April	15	Saturday	2
10.75	5	0	3.5475	2023	2	4	April	18	Tuesday	2
13.12	4	0	3.8048	2023	2	4	April	18	Tuesday	2
11.62	2	0	3.6022	2023	2	4	April	18	Tuesday	2
2.88	1	0	1.4112	2023	3	7	July	9	Sunday	3
23.92	4	0	4.0664	2023	3	7	July	9	Sunday	3
141.4	5	0	38.178	2023	4	10	October	28	Saturday	4
480.74	2	0	14.4222	2023	4	10	October	28	Saturday	4
6.28	2	0	0.062800000000000002	2023	4	10	October	28	Saturday	4
12.39	3	0	3.717	2023	4	10	October	31	Tuesday	4
11.34	3	0	5.2164	2023	4	10	October	31	Tuesday	4
64.74	13	0	30.4278	2023	4	10	October	31	Tuesday	4
405.64	4	0	12.1692	2023	4	10	October	31	Tuesday	4
80.3	5	0	20.878	2023	4	10	October	31	Tuesday	4
251.91	9	0	47.8629	2023	4	10	October	31	Tuesday	4
44.4	3	0	22.2	2023	4	11	November	1	Wednesday	4
21.4	5	0	9.63	2023	4	11	November	1	Wednesday	4
6.72	4	0	3.36	2023	4	11	November	3	Friday	4

Figure 11: Superstore Sales table verified in Power BI Model View

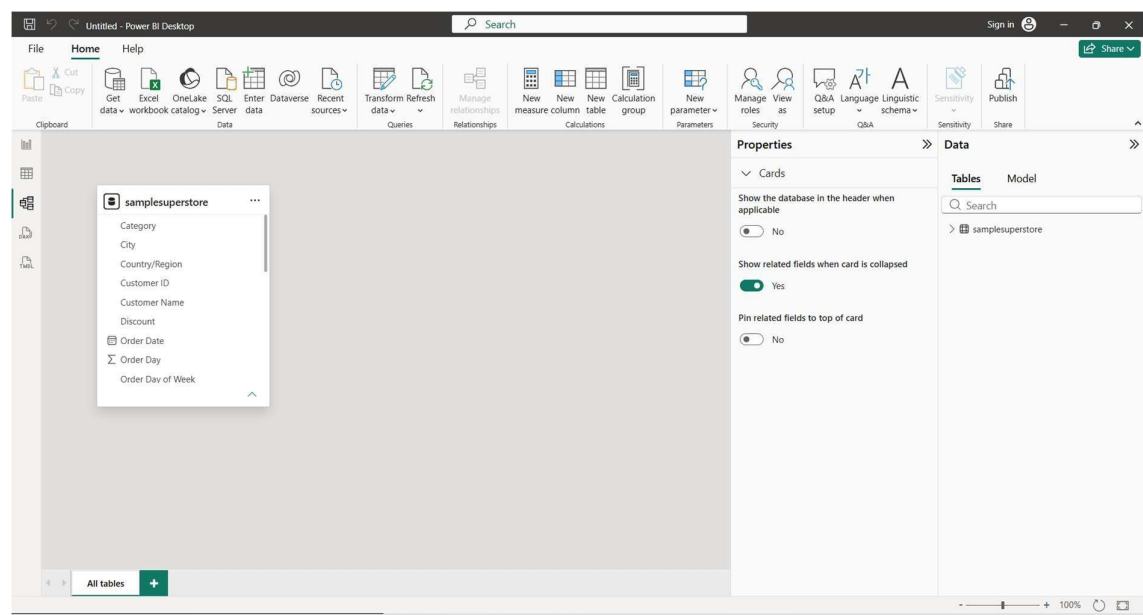


Figure 12: Order Month configured for chronological sorting using Order Month Number

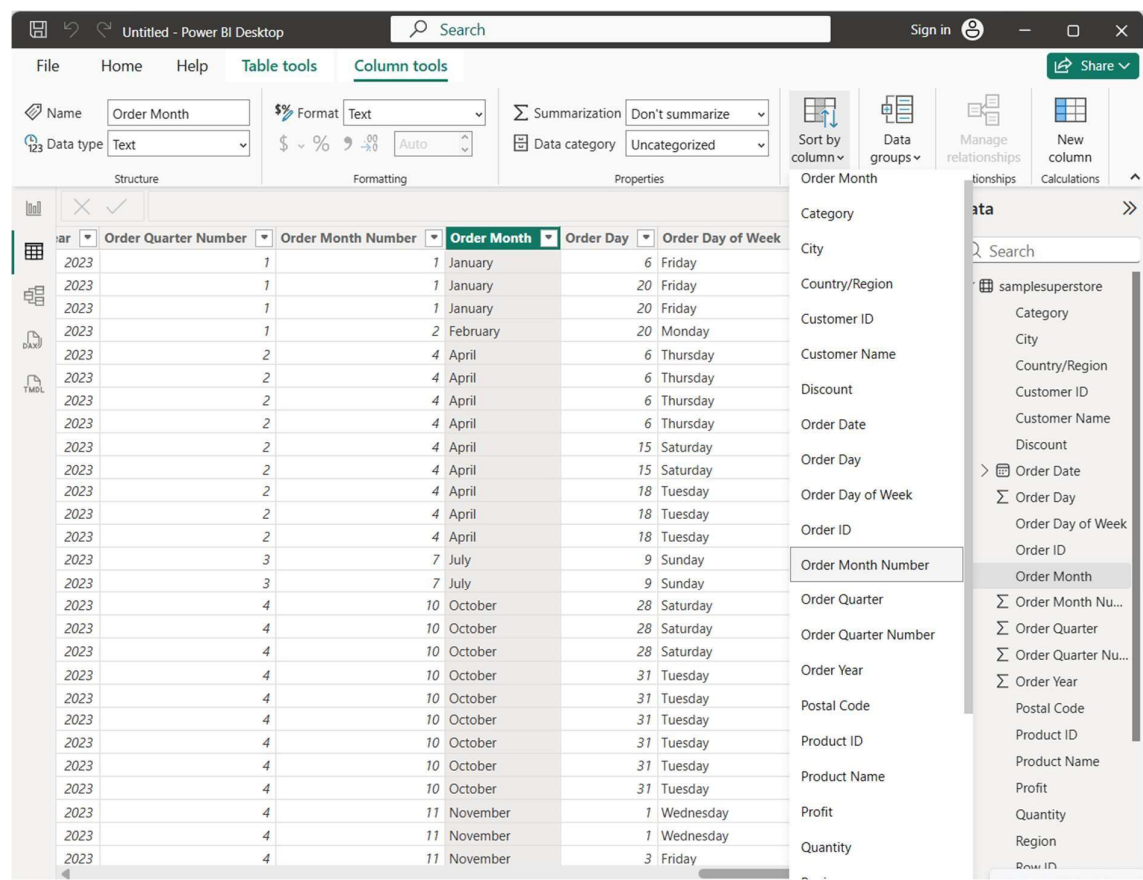


Figure 13: Order Quarter configured for chronological sorting using Order Quarter Number

Year	Order Quarter Number	Order Month Number	Order Month	Order Day	Order Day of Week	Order Quarter
2023	1	1	January	6	Friday	1
2023	1	1	January	20	Friday	1
2023	1	1	January	20	Friday	1
2023	1	2	February	20	Monday	1
2023	2	4	April	6	Thursday	2
2023	2	4	April	6	Thursday	2
2023	2	4	April	6	Thursday	2
2023	2	4	April	6	Thursday	2
2023	2	4	April	15	Saturday	2
2023	2	4	April	15	Saturday	2
2023	2	4	April	18	Tuesday	2
2023	2	4	April	18	Tuesday	2
2023	2	4	April	18	Tuesday	2
2023	3	7	July	9	Sunday	3
2023	3	7	July	9	Sunday	3
2023	4	10	October	28	Saturday	4
2023	4	10	October	28	Saturday	4
2023	4	10	October	28	Saturday	4
2023	4	10	October	31	Tuesday	4
2023	4	10	October	31	Tuesday	4
2023	4	10	October	31	Tuesday	4
2023	4	10	October	31	Tuesday	4
2023	4	10	October	31	Tuesday	4
2023	4	10	October	31	Tuesday	4
2023	4	11	November	1	Wednesday	4
2023	4	11	November	1	Wednesday	4
2023	4	11	November	3	Friday	4

Table: samplesuperstore (10,194 rows) Column: Order Quarter (4 distinct values)

Conclusion

The Superstore Sales dataset was successfully imported into Power BI and prepared for analysis using Power Query Editor. The data types were verified, date fields were transformed, text data was cleaned, and seven useful date attributes were created from the Order Date column. The transformed dataset was then loaded into Power BI and verified in Data View and Model View. Chronological sorting was also configured for Order Month and Order Quarter. Overall, the Week 1 task successfully prepared a clean and structured dataset for data modeling, DAX calculations, and dashboard development in Week 2.

